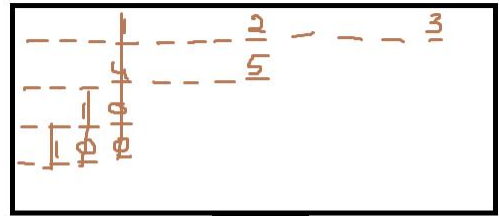
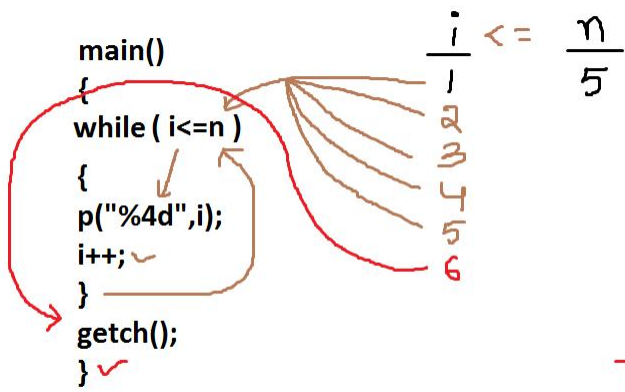


Printing 1..n natural no's:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 10 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1,n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)
{
printf("%4d",i);
i++;
}
getch();
}
```

Enter n value 100

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38
41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98



$$\begin{array}{r} 1 \\ 10 \\ 100 \\ \hline 300 \end{array}$$

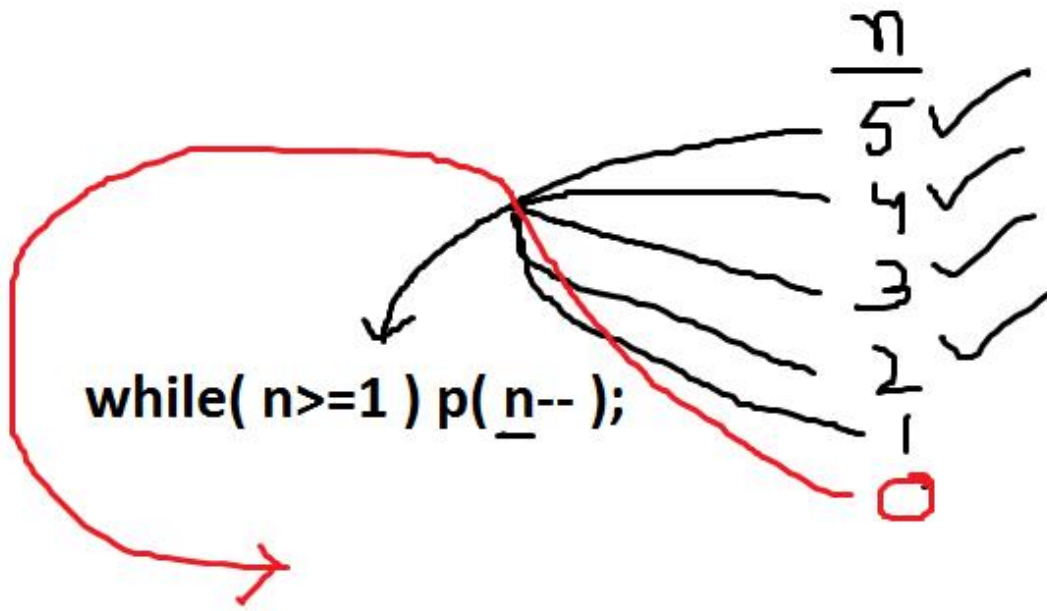
$$\begin{array}{r} 1 \\ 10 \\ 100 \\ \hline 111 \end{array}$$

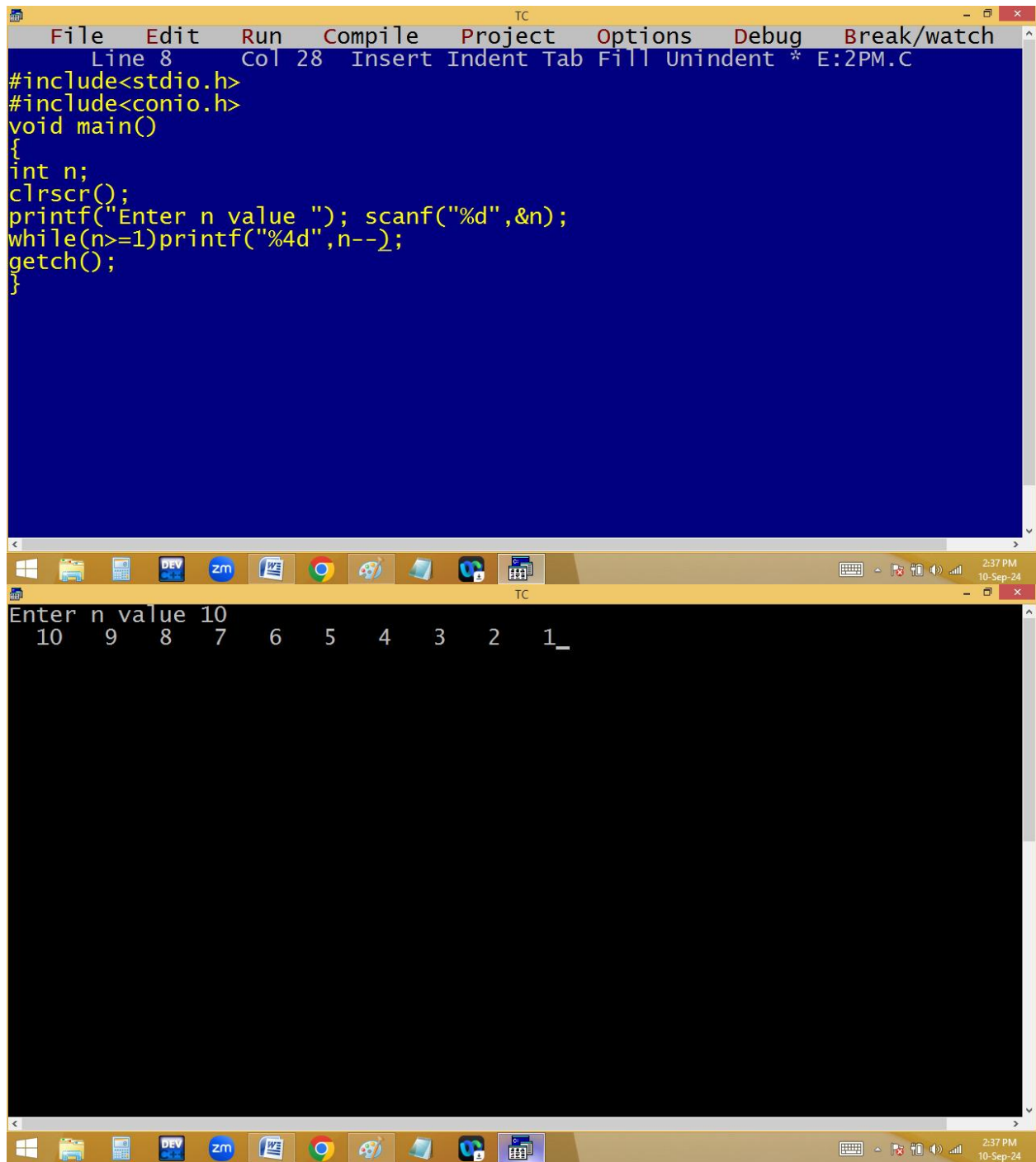
The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a C program. The code includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside `main`, it declares an integer `i` and takes an input `n`. It then uses a `while` loop to print numbers from `n` down to 1, with a `printf` statement for each iteration. The bottom window shows the program's execution. It prompts the user to "Enter n value" and the user has entered "10". The output shows the numbers 1 through 10 printed in reverse order, with each number on a new line. The Windows taskbar at the bottom indicates the time is 2:34 PM on 10-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1,n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)printf("%4d",i++);
getch();
}
```

Enter n value 10
1 2 3 4 5 6 7 8 9 10_

Printing 1..n numbers in reverse order:

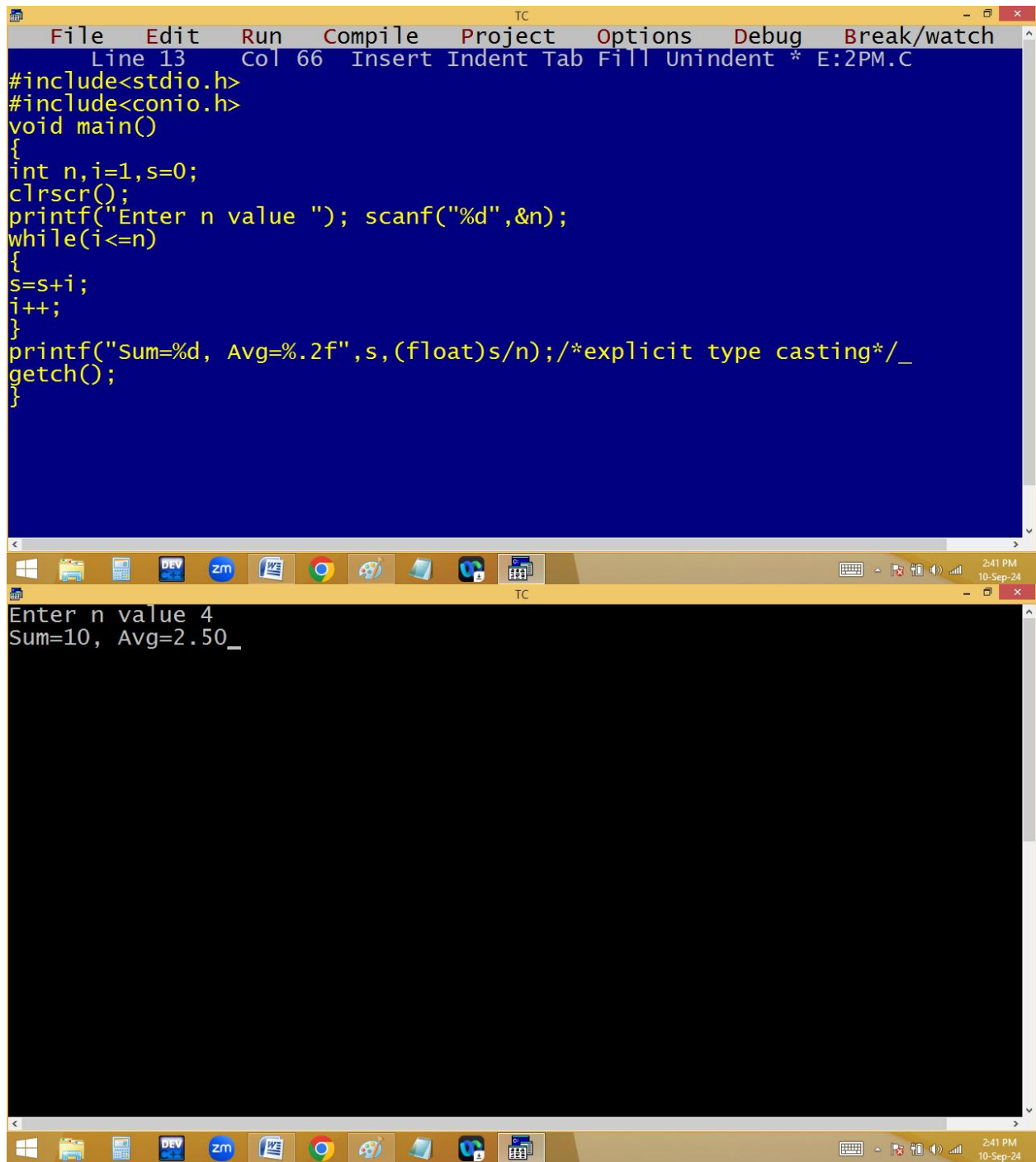




```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 28 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>=1)printf("%4d",n--);
getch();
}
```

Enter n value 10
10 9 8 7 6 5 4 3 2 1_

Finding 1..n numbers sum and avg[mean]:



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a C program named E:2PM.C. The code calculates the sum and average of the first 'n' natural numbers. The code is as follows:

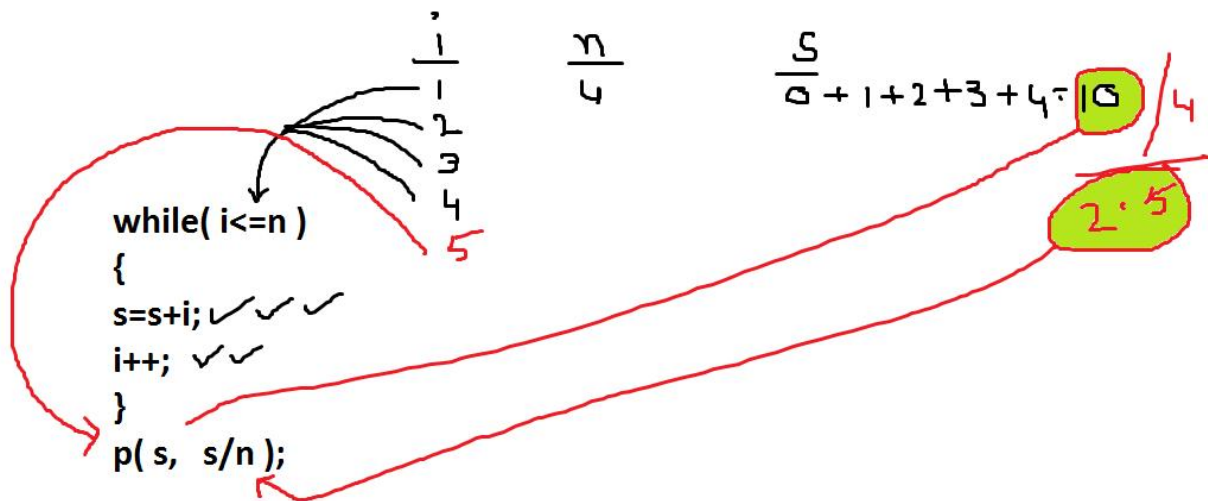
```
Line 13 Col 66 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i=1,s=0;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)
{
s=s+i;
i++;
}
printf("Sum=%d, Avg=%.2f",s,(float)s/n);/*explicit type casting*/_
getch();
}
```

The bottom window shows the program's execution. It prompts the user to enter a value for 'n'. The user has entered '4'. The program outputs the sum as 10 and the average as 2.50.

```
Enter n value 4
Sum=10, Avg=2.50_
```

The Windows taskbar at the bottom shows the time as 2:41 PM on 10-Sep-24. The taskbar includes icons for the Start menu, File Explorer, DEV, zm, WE, Chrome, and other applications.

```
Enter n value 10
Sum=55, Avg=5.50
```



Without using loop:

$n=4$

$$s = n \cdot (n+1) / 2 \rightarrow 4 \cdot (4+1) / 2 \rightarrow 4 \cdot 5 / 2 \rightarrow 20 / 2 = 10$$


```
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 15 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,s;
clrscr();
printf("Enter n value "); scanf("%d",&n);
s = n*(n+1)/2;
printf("Sum=%d, Avg=%.2f",s,(float)s/n);/*explicit type casting*/
getch();
}
```

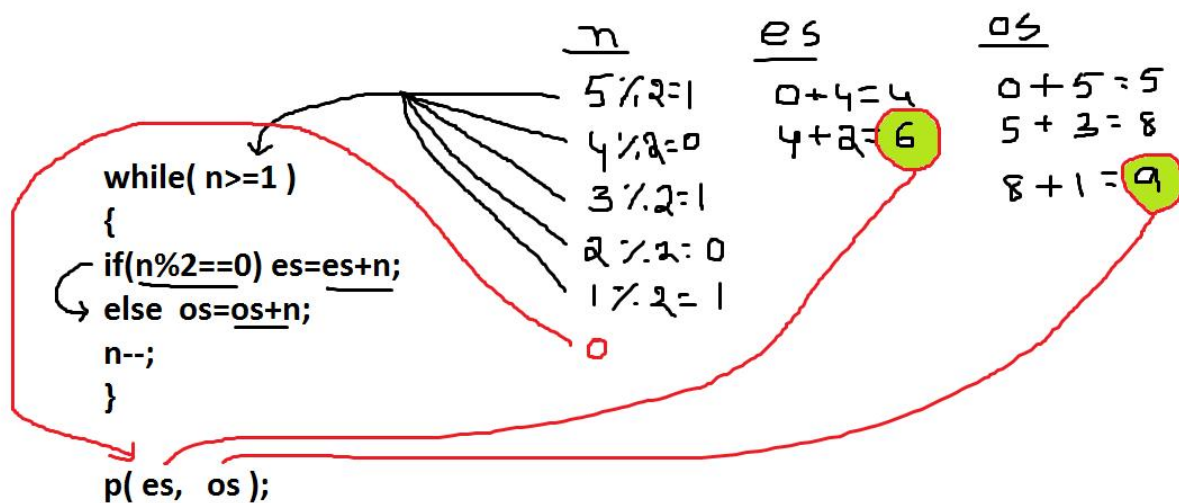
Enter n value 5
Sum=15, Avg=3.00

Finding 1..n even, odd numbers sum.

$$5 \left\{ \begin{array}{l} 2 + 4 = 6 \\ 1 + 3 + 5 = 9 \end{array} \right.$$

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 1 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,es=0,os=0;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>=1)
{
if(n%2==0)es+=n; /*es=es+n;*/
else os+=n; /* os=os+n; */
n--;
}
printf("Even Sum=%d, Odd Sum=%d",es,os);
getch();
}
```

```
Enter n value 5
Even Sum=6, Odd Sum=9_
```



Factorial of given no:

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a C program that calculates the factorial of a number. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,f=1;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>1)
{
f=f*n;
n--;
}
printf("Factorial=%d",f);
getch();
}
```

The bottom screenshot shows the execution of the program. The user has entered the value 4, and the program has calculated the factorial, which is 24.

```
Enter n value 4
Factorial=24
```

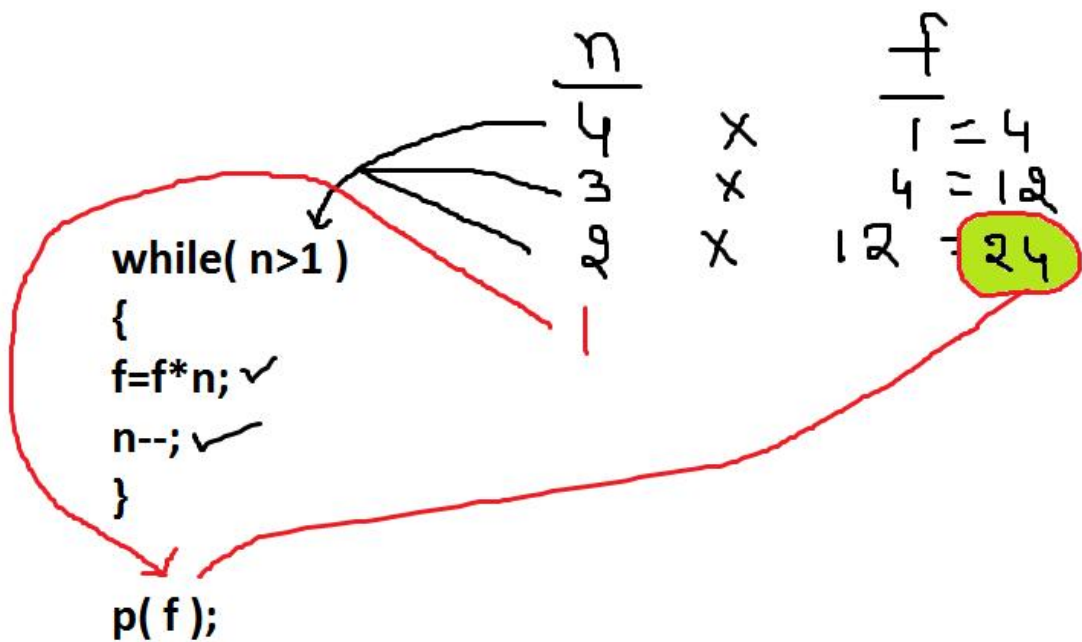
The Windows taskbar at the bottom of both screenshots shows the time as 2:54 PM on 10-Sep-24. The taskbar includes icons for the Start menu, File Explorer, DEV, zm, WE, Chrome, Paint, and the TC application.

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the output of a program: "Enter n value 8" followed by "Factorial=-25216". The bottom window shows the source code for the program, which is a C program to calculate the factorial of a number. The code includes headers for stdio.h and conio.h, and uses a while loop to calculate the factorial. The IDE's menu bar and taskbar are also visible.

```
Enter n value 8
Factorial=-25216
```

```
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 21 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n; long f=1;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>1)
{
f=f*n;
n--;
}
printf("Factorial=%ld",f);
getch();
}
```

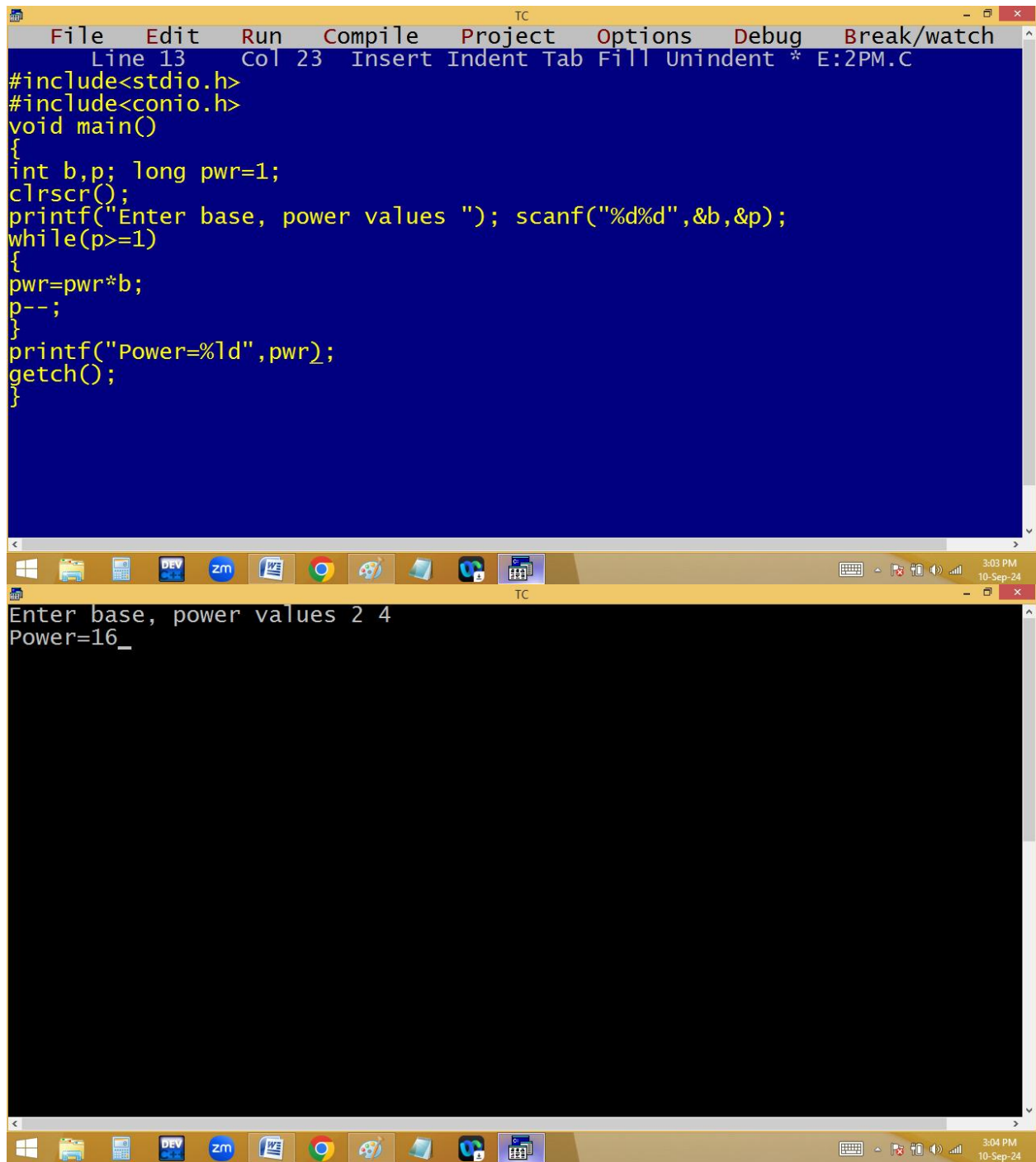
```
Enter n value 8
Factorial=40320
```



Finding power using user defined program

b / $2^4 = 16$

$$2 \times 2 \times 2 \times 2 = 16$$



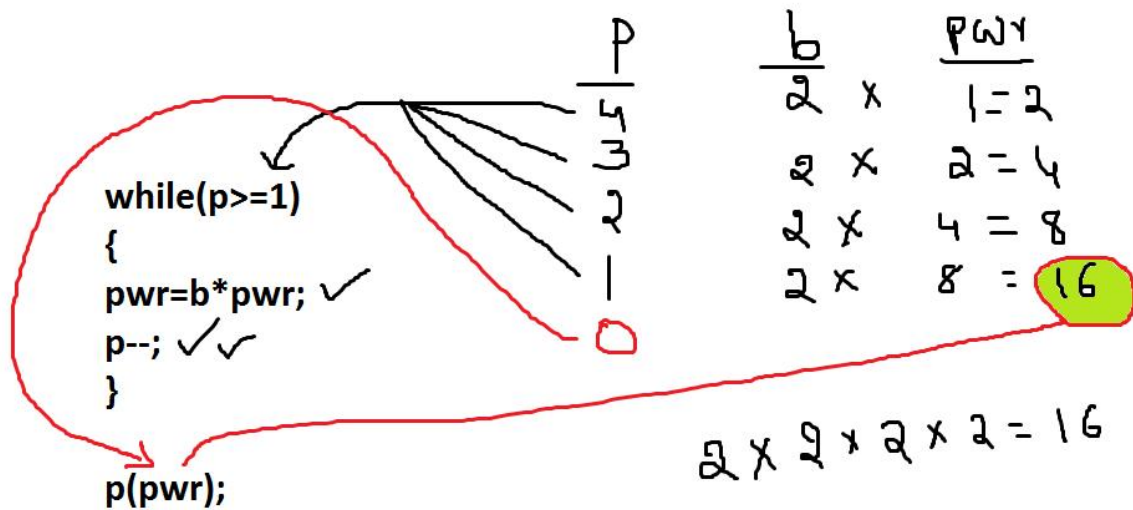
The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a C program named E:2PM.C. The code uses `#include<stdio.h>` and `#include<conio.h>`. The `main` function declares `int b,p;` and `long pwr=1;`, then calls `clrscr();`. It prompts the user with `printf("Enter base, power values ");` and reads input with `scanf("%d%d",&b,&p);`. A `while(p>=1)` loop calculates the power by multiplying `pwr` by `b` and decrementing `p` until `p` is 0. Finally, it prints the result with `printf("Power=%ld",pwr);` and waits for a key press with `getch();`.

The bottom window shows the program's execution. It displays the prompt `Enter base, power values` followed by the user input `2 4`. The output is `Power=16_`, where the underscore indicates the cursor position after the program has finished execution.

```
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 23 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int b,p; long pwr=1;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
while(p>=1)
{
pwr=pwr*b;
p--;
}
printf("Power=%ld",pwr);
getch();
}
```

Enter base, power values 2 4
Power=16_

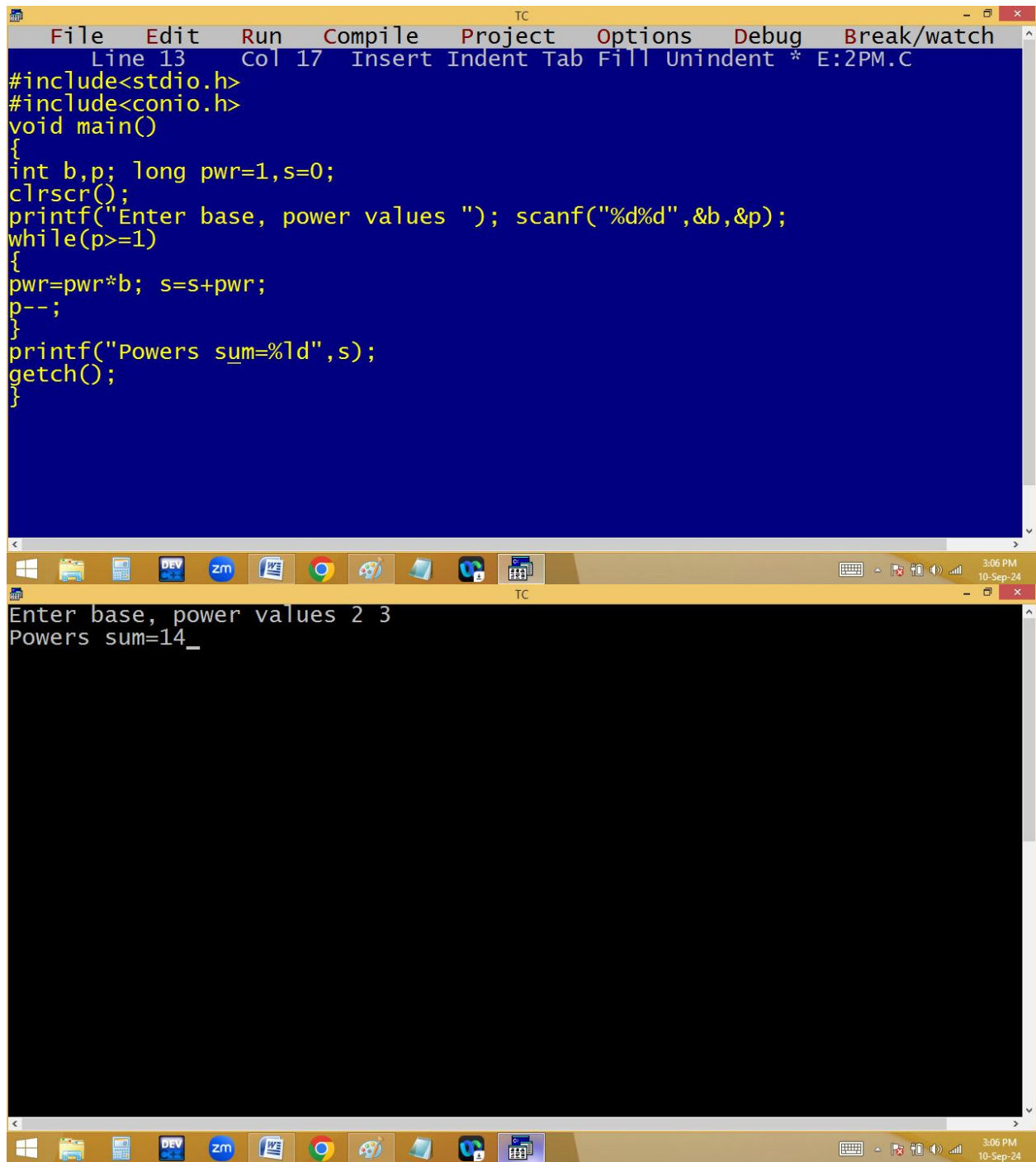

```
TC
Enter base, power values 100 3
Power=1000000_
```



find the powers sum:

$$2^3 = 2^1 + 2^2 + 2^3$$

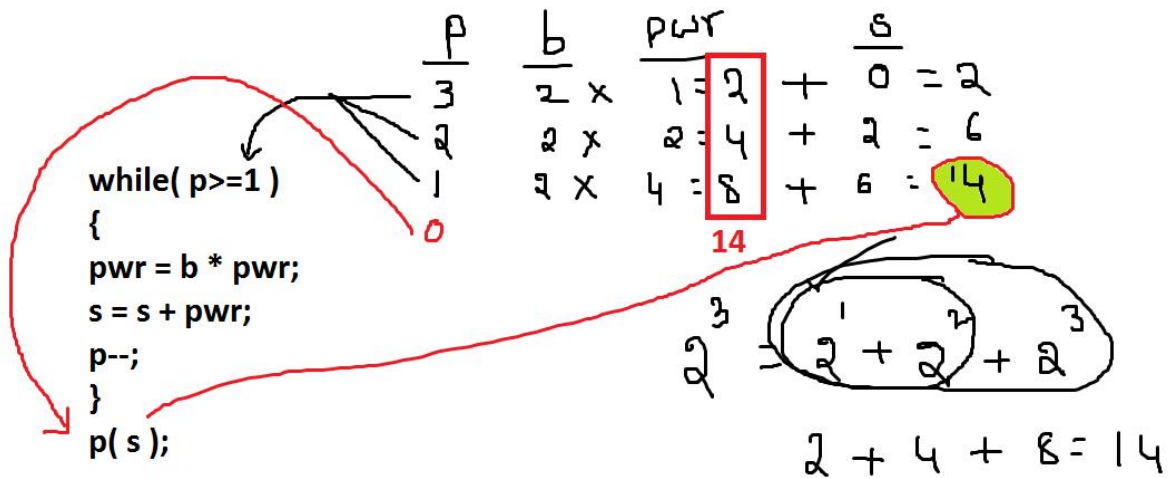
$$2 + 4 + 8 = 14$$



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 17 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int b,p; long pwr=1,s=0;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
while(p>=1)
{
pwr=pwr*b; s=s+pwr;
p--;
}
printf("Powers sum=%ld",s);
getch();
}
```

Enter base, power values 2 3
Powers sum=14_

TC



Using pwr():

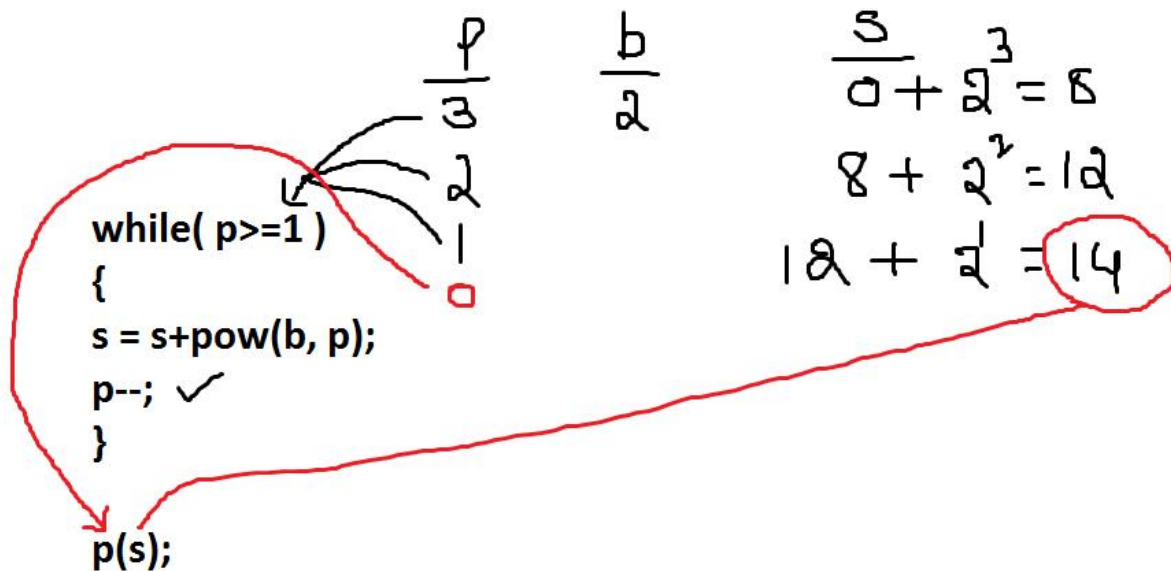
```

while( p >= 1 )
{
    s = s + pow(b, p);
    p--;
}

p(s);
  
```

2, 3 = 8
2, 2 = 4
2, 1 = 2

$\frac{p}{3}$	$\frac{b}{2}$	$\frac{s}{0}$
		8
		4
		<u>2</u>
		14

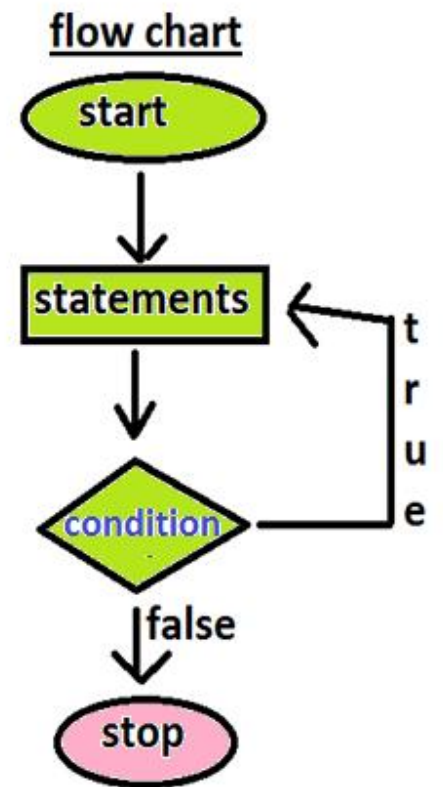
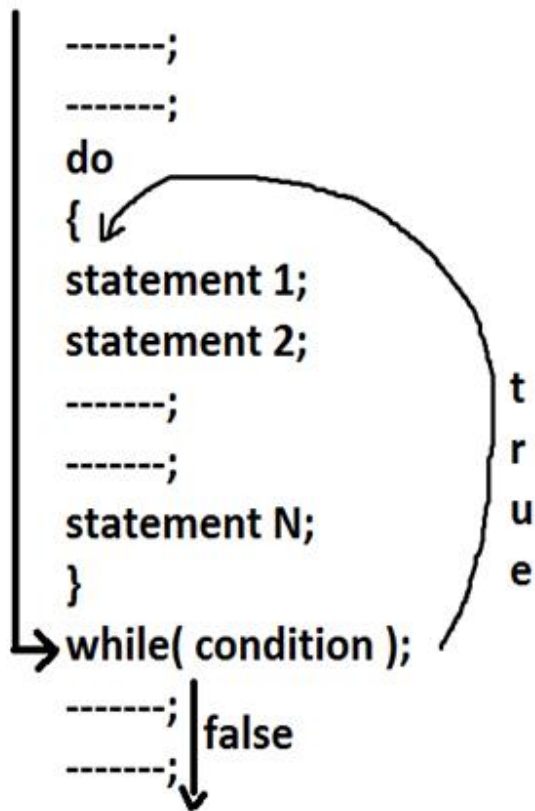


do .. while:

- It is an exit control loop. i.e. in a do while the condition is tested at last.
- Here do , while are the keywords.
- It is also used to repeat a program several times based on a condition.
- In a do while, do block statements are executed first and later while condition is tested. If the while condition is true then once again the do block

statements are repeated. Like this the process is continued until the while condition becomes false.

- In do while, the while should be end with semicolon (;) .
- Regardless of while condition, the do statements are executed at least one time. Due to this sometimes we are getting unwanted results [garbage values].
- Use do while whenever it is compulsory because of in do while the program is controlled at the bottom / last.



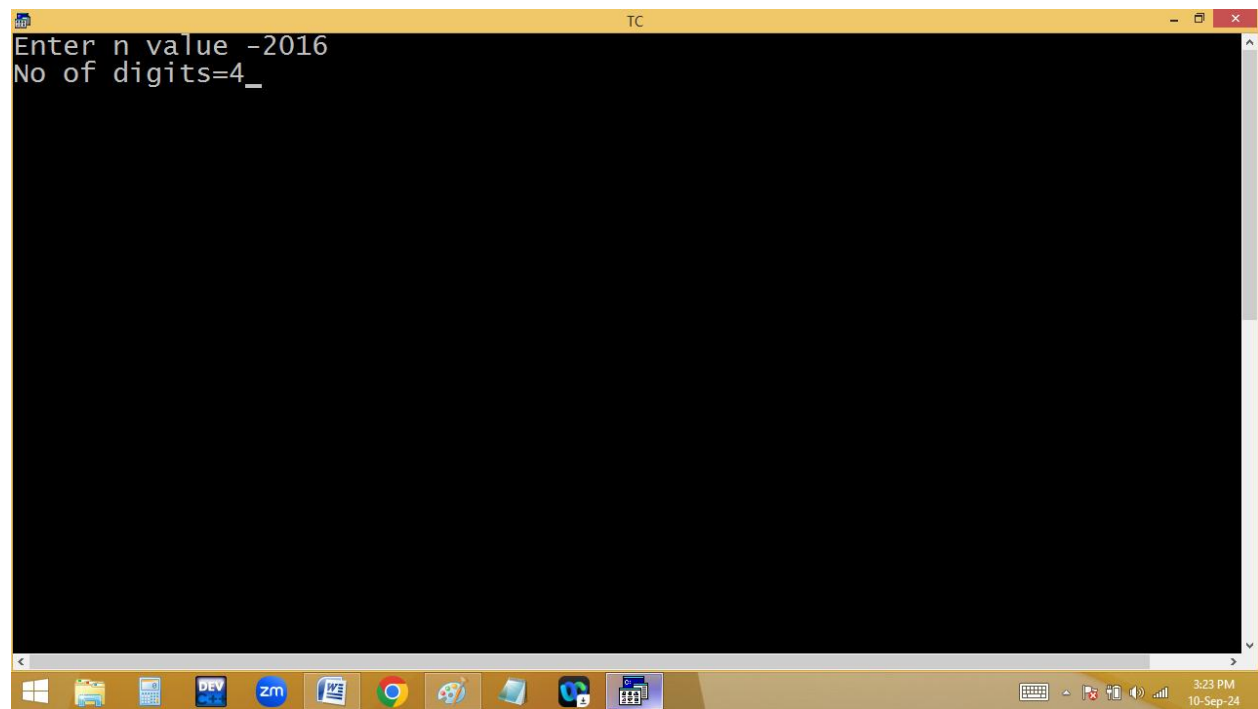
Finding no of digits in given no.

2016 → 4 digit no

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a C program named E:2PM.C. The code includes headers for stdio.h, conio.h, and math.h. It defines a main function where a long integer n is declared and initialized to 0. The screen is cleared using clrscr(). A printf statement prompts the user to enter an n value, followed by a scanf statement to read the input. A while loop then iterates as long as n is not zero, incrementing a counter c and dividing n by 10 in each iteration. After the loop, a printf statement displays the number of digits, and getch() is used to pause the program. The bottom screenshot shows the program's execution. It prompts for an input value, which is 2016. The program then outputs that the number of digits is 4. The Windows taskbar at the bottom of both screenshots shows the time as 3:22 PM on 10-Sep-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 10 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
long n; int c=0;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
while(n!=0)
{
c++;
n=n/10;
}
printf("No of digits=%d",c);
getch();
}
```

Enter n value 2016
No of digits=4



```
TC
Enter n value 9
No of digits=1
```

```
TC
Enter n value 0
No of digits=0
```

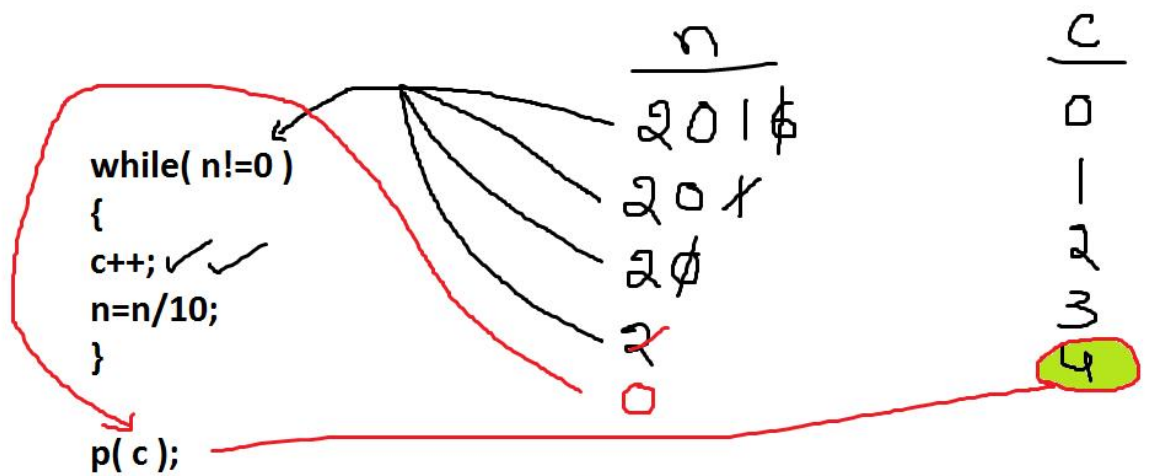
The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a C program named E:2PM.C. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 3 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int c=0;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
do
{
c++;
n=n/10;
}
while(n!=0);
printf("No of digits=%d",c);
getch();
}
```

The bottom window shows the execution output of the program. It displays the prompt "Enter n value 0" and the result "No of digits=1". The Windows taskbar at the bottom indicates the time is 3:25 PM on 10-Sep-24.

```
TC
Enter n value 123
No of digits=3_
```

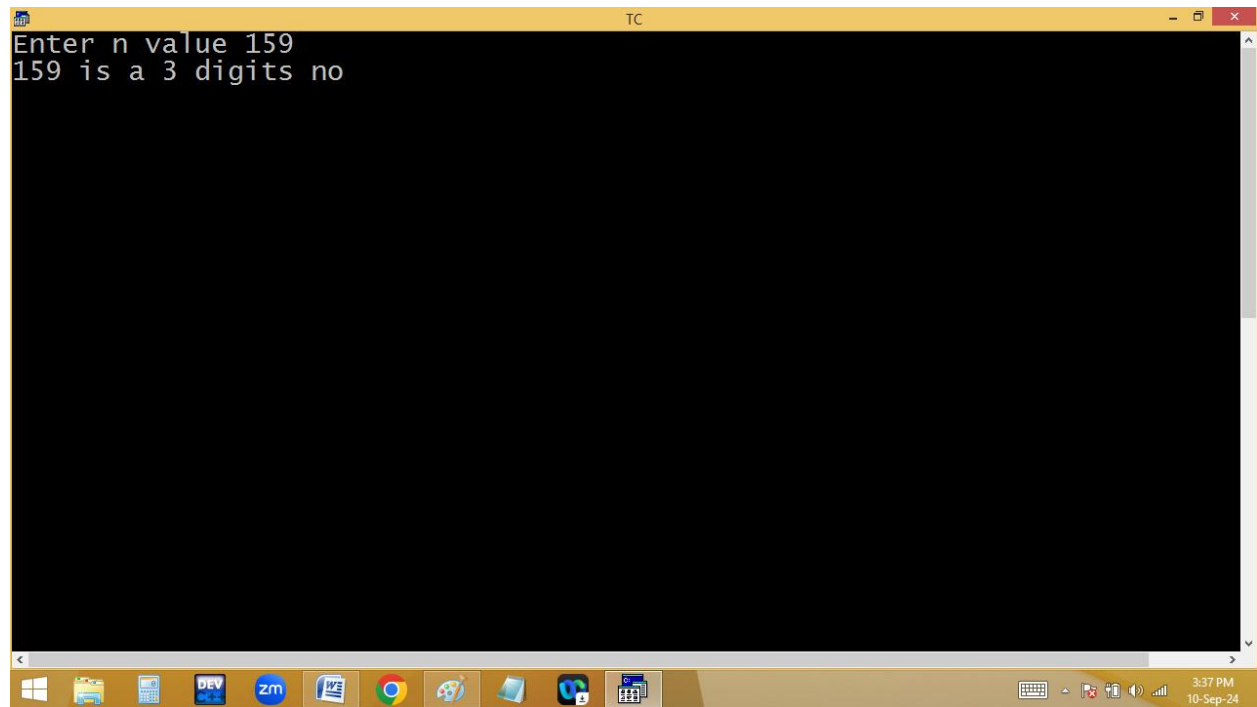
```
TC
Enter n value -234
No of digits=3_
```



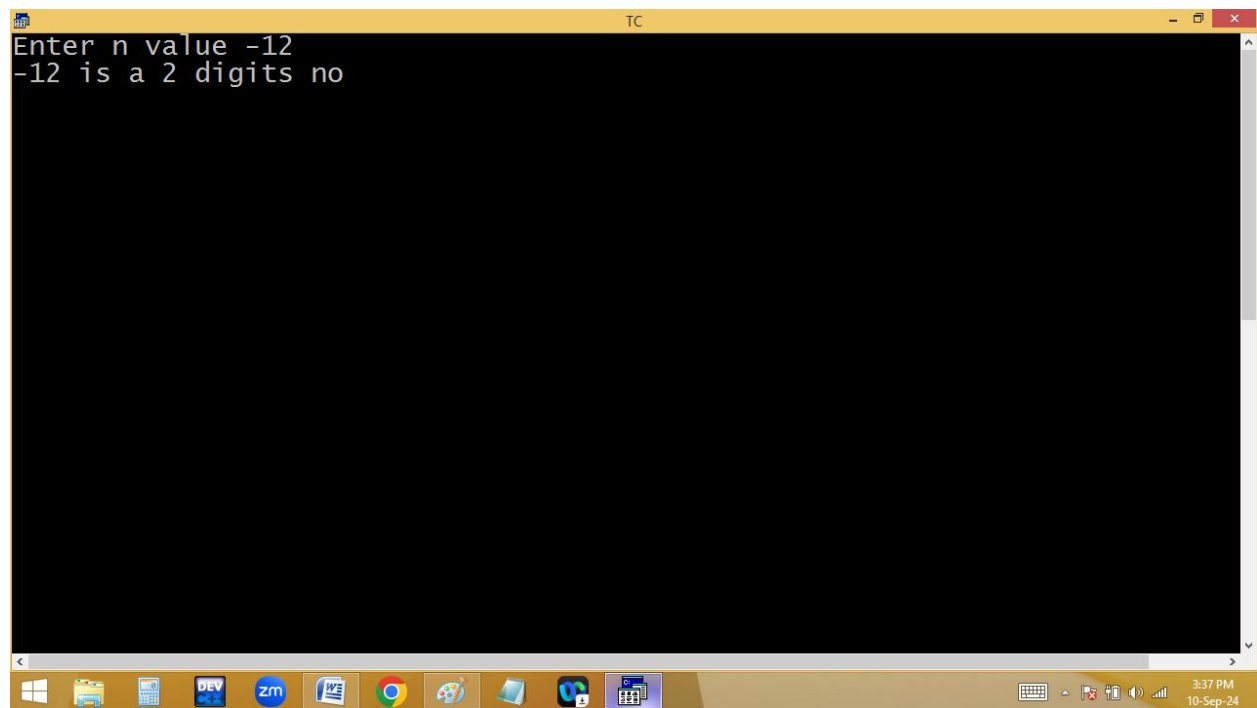
Without using loop / goto label:

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int c;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
c=printf("%ld",n);
if(n<0) c--;
printf(" is a %d digits no",c);
getch();
}
```

```
TC
Enter n value 159
159 is a 3 digits no
```



```
TC
Enter n value -12
-12 is a 2 digits no
```



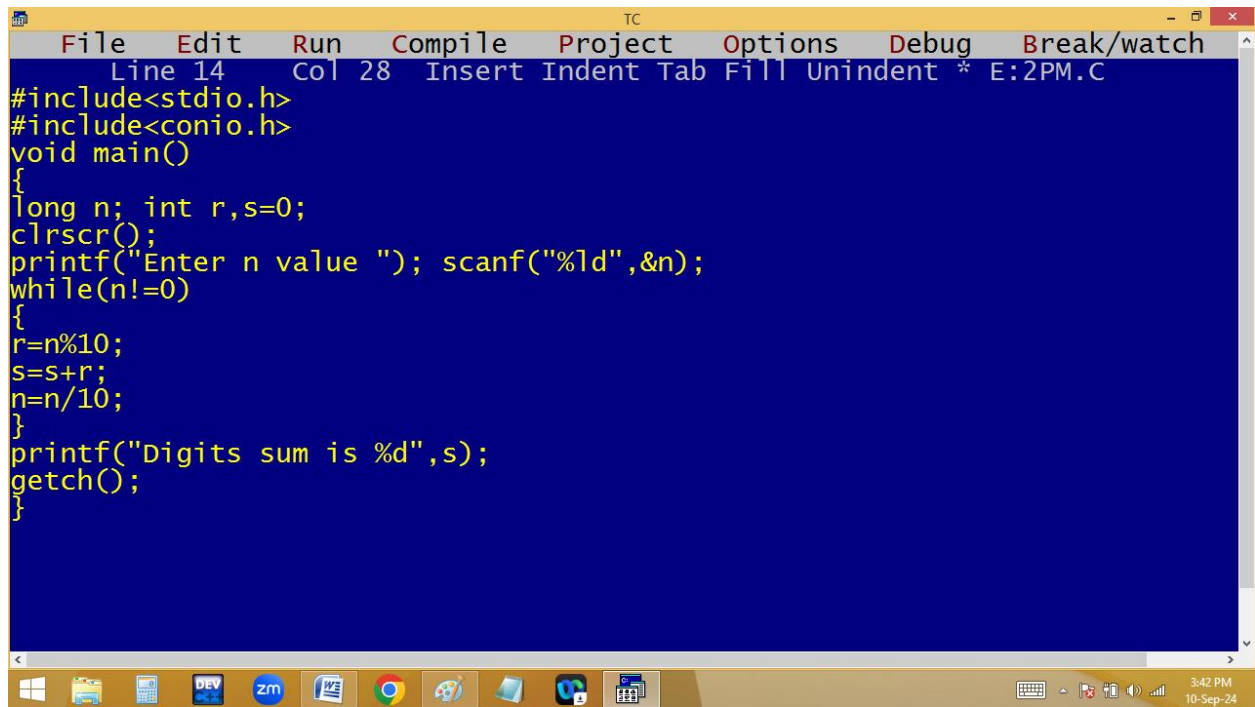
```
Enter n value 0
0 is a 1 digits no
```

175 is a 3 digits no

$\frac{n}{175}$
 $c = p(\text{"\%ld", } n);$
 $p(\text{" is a \%d digits no", } c);$

Finding digits sum:

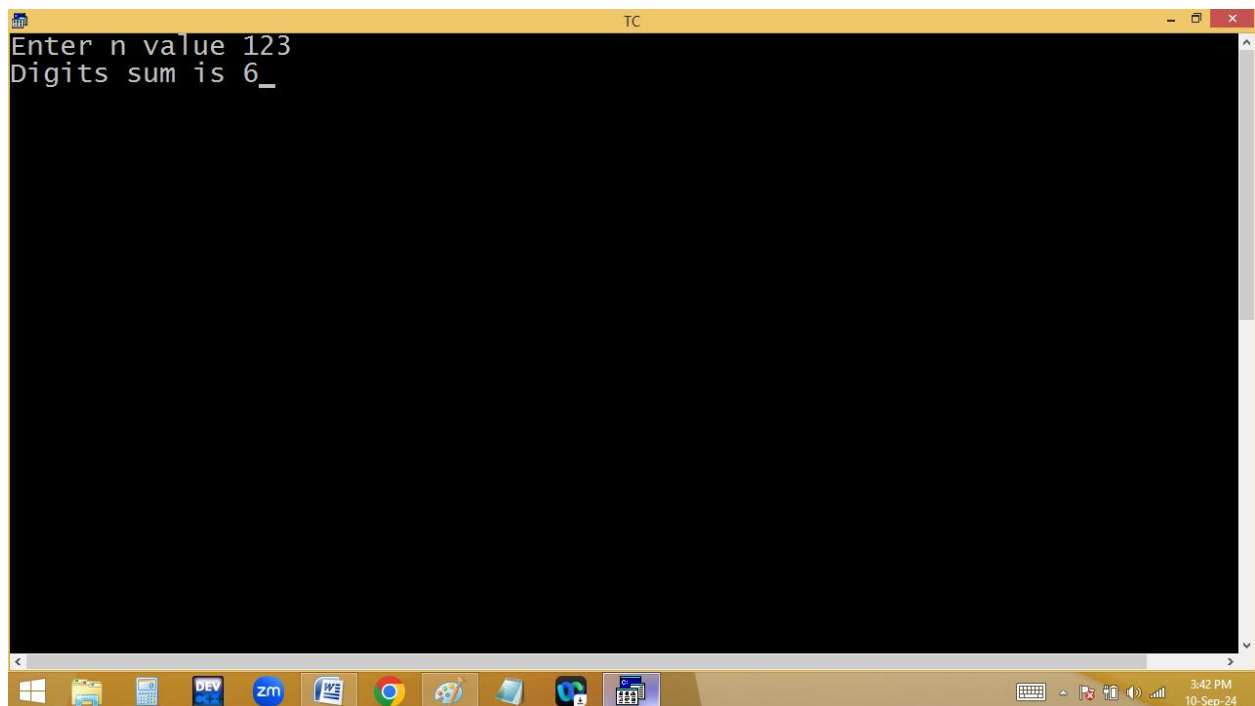
294 $\rightarrow 2+9+4=15$



The screenshot shows the Turbo C++ (TC) editor window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 14 Col 28 Insert Indent Tab Fill Unindent * E:2PM.C'. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n; int r,s=0;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    while(n!=0)
    {
        r=n%10;
        s=s+r;
        n=n/10;
    }
    printf("Digits sum is %d",s);
    getch();
}
```

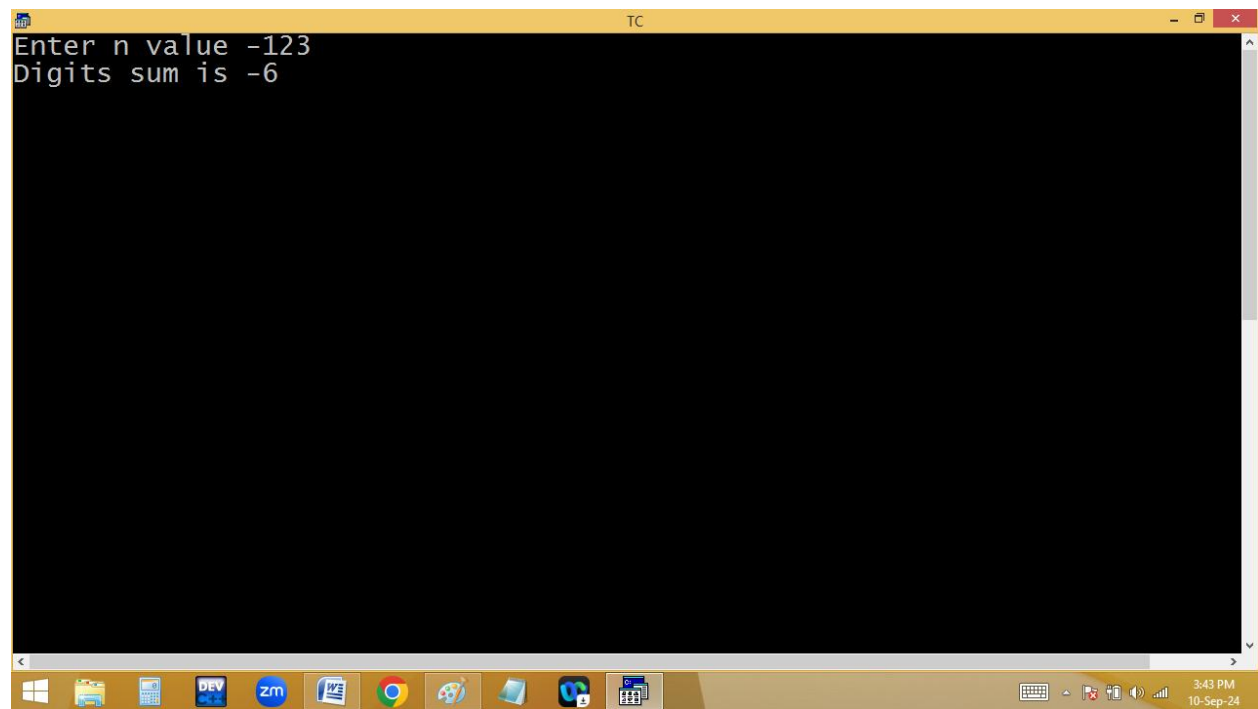
The Windows taskbar at the bottom shows various icons including Windows, File Explorer, DEV, zm, Notepad, Chrome, Paint, and the TC application icon. The system clock shows 3:42 PM on 10-Sep-24.



The screenshot shows the Turbo C++ (TC) editor window displaying the output of the program. The text 'Enter n value 123' is on the first line, and 'Digits sum is 6_' is on the second line. The rest of the editor window is black.

```
Enter n value 123
Digits sum is 6_
```

The Windows taskbar at the bottom is identical to the first screenshot, showing the same icons and system clock (3:42 PM on 10-Sep-24).



A screenshot of a Windows 10 desktop environment. A terminal window titled "TC" is open, displaying the output of a program. The text in the terminal is:

```
Enter n value -123
Digits sum is -6
```

The desktop background is a solid light blue. The taskbar at the bottom contains several icons: the Windows Start button, File Explorer, Microsoft Edge, a blue icon with "DEV" text, a blue icon with "zm" text, a Word document icon, the Google Chrome logo, a game controller icon, a folder icon, a blue icon with a white "S", and a calendar icon. On the right side of the taskbar, there are system tray icons for keyboard, network, and volume, along with the system clock showing "3:43 PM" and "10-Sep-24".

```
Enter n value 0
Digits sum is 0_
```

	<u>n</u>	<u>r</u>	<u>s</u>
	294	4	0
	29	9	4
	2	2	13
	0	0	15

`while(n!=0)`
{
 `r=n%10;`
 `s=s+r;` ✓
 `n=n/10;` ✓
}
`p(s);`

Any `no%10` gives last digit.
Any `no/10` removes last digit.

Home work:

19357 → finding 1 and last digits of given no → 1 first, 7 last digit

19357 → finding 5 position → 4

1920 → finding max, min digits → max 9, min 0

